Type : Research Based

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A Vision–Language System for Intelligent Question Answering and Information Extraction from Financial Documents

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Introduction

Vision–Language and Intelligent Learning focuses on building systems that can jointly understand visual information and natural language. In many real-world scenarios, information is not present as plain text but is embedded within images, layouts, tables, and structured documents.

Financial documents such as invoices, bank statements, receipts, and tax forms are visually complex and vary widely in format. Traditional OCR-based systems can extract text but often fail to capture document layout, context, and relationships between fields, leading to incomplete or incorrect understanding.

Recent advances in Vision–Language Models and layout-aware transformers have enabled Document Visual Question Answering (DocVQA), where systems can directly answer natural language queries from document images. In the financial domain, where accuracy and reliability are critical, such intelligent document understanding systems help reduce errors, manual effort, and operational risk.

Problem Statement & Motivation

Problem Statement "To design an intelligent Document VQA system that can effectively understand financial documents and accurately answer user queries despite variations in layout and format."

- Financial institutions handle large volumes of documents such as invoices, bank statements, and receipts that contain important information for decision-making and record keeping. These documents are often unstructured and vary in layout, format, and quality, making automatic understanding difficult. Existing Document VQA models such as LayoutLM, LayoutLMv2, OCR-based systems, and Donut combine visual and textual features but struggle with complex layouts and real- world document variations, leading to reduced accuracy and the need for manual verification.

Motivation

- Accurate and reliable information extraction is crucial in financial applications. Current Document VQA systems often fail to handle complex document structures and provide answers without sufficient reliability, increasing manual effort.
- This motivates the development of a more robust Document VQA system that can better understand financial documents, improve answer accuracy, and reduce human intervention, making document processing more efficient and trustworthy.

Literature Survey (1–5)

No	Author/Year	Title	Research Focus	Limitations	Research Gap
1	Li et al., 2025	GenRecal	Reasoning transfer from large VLMs	Synthetic reasoning limits real-world use	Need financial DocVQA systems
2	Wu et al., 2023	VQA Survey	Comparative study of OCR and VLM methods	No experiments; rapidly evolving field	Need domain-specific benchmarks
3	Kim et al., 2022	Donut	OCR-free document understanding	Requires large training data	Robust layout understanding
4	Xu et al., 2022	LayoutLMv3	Multimodal learning (text+image+layout)	OCR dependent and costly	Efficient OCR-free models
5	Wang et al., 2021	LaTr	Layout-aware reasoning using spatial features	OCR bias; scene-text focus	Financial layout reasoning models

Table 1.1: Literature Survey (1–5)

Literature Survey (6–10)

No	Author/Year	Title	Research Focus	Limitations	Research Gap
6	Choi et al., 2025	FinAgentBench	Multi-step financial reasoning	Weak retrieval	Better reasoning integration
7	Jin et al., 2025	OCR+VLM	Structured extraction	Dataset dependency	Robust DocVQA
8	Choi et al., 2025	FinDER	Retrieval QA	Ambiguous queries	Improved retrieval
9	Xue et al., 2024	FAMMA	Multimodal QA benchmark	Low accuracy	Strong QA systems
10	Reddy et al., 2024	DocFinQA	Long-document QA	Weak long reasoning	Better reasoning models

Table 1.2: Literature Survey (6–10)

Literature Survey (11–15)

No	Author/Year	Method	Research Focus	Limitations	Research Gap
11	Xichao Jin et al., 2024	Graph-based reasoning (GCNs)	Numerical reasoning in visual documents	High computation; OCR dependency	Hybrid LLM + graph models
12	Reddy et al., 2024	LLM Evaluation	Financial numerical reasoning	High cost	Efficient symbolic reasoning
13	Qingyi Tao et al., 2022	Multimodal Transformer	Visual QA reasoning	OCR noise	Semantic reasoning
14	Zhengnan Xue et al., 2021	Retriever–Generator	Table reasoning	Long chains fail	Domain-aware reasoning
15	Mathew et al., 2021	OCR QA	Document QA baseline	Poor layout	Layout-aware models

Table 1.3: Literature Survey (11–15)

Literature Survey (16–20)

No	Author/Year	Method	Research Focus	Limitations	Research Gap
16	Chen et al., 2025	Multi-agent LLM	Multilingual document QA	High computation	Real-world systems
17	Zhu et al., 2024	Table QA	Financial tables	Limited scope	Unstructured QA
18	Xue et al., 2024	Multimodal QA	Financial QA systems	Limited diversity	Complex reasoning
19	Mathew et al., 2022	Chart QA	Infographic QA	Sparse text	Dense-text reasoning
20	Mathew et al., 2021	Visual QA	Document QA	Weak multi-page	Long-context reasoning

Table 1.4: Literature Survey (16–20)

Literature Survey (21–25)

No	Author/Year	Title	Research Focus	Limitations	Research Gap
21	Benchea et al., 2025	DIVE-Doc	OCR-free DocVQA	Limited evaluation	Hybrid models
22	Mohammadshirazi et al., 2025	DLaVA	OCR-free QA	Alignment issues	Grounded reasoning
23	Nguyen et al., 2025	DocMIA	Privacy analysis	No defense	Secure QA
24	Mohammadshirazi et al., 2025	DocVAL	Spatial grounding	High cost	Domain adaptation
25	Van Landeghem et al., 2023	DUDE	Long document QA	OCR dependent	Financial reasoning

Table 1.5: Literature Survey (21–25)

Novelty of the Work & Research Gap Analysis

Novelty of the work

- Proposes an integrated vision-language framework for intelligent question answering and information extraction from financial documents.
- Uses a single transformer-based architecture instead of separate OCR, layout, and language pipelines.
- Combines visual, spatial, and textual features in one unified model.
- Incorporates layout-aware reasoning to understand structured documents like invoices and balance sheets.
- Provides explainable question answering with confidence scores.
- Enhances transparency, reliability, and real-world financial decision support.

Research gap Analysis

- No unified end-to-end financial document reasoning system
- Weak long-document global multi-hop reasoning
- OCR-fragile pipelines with poor robustness
- Inadequate numerical and structured financial reasoning
- Lack of explainability and evidence-grounded reasoning
- Limited multilingual and real-world generalization
- Poor scalability and high computational cost

Proposed System and Use Case

We propose a layout-aware Vision-Language DocVQA system designed for financial documents.

Key Features:

- Unified transformer-based architecture
- Combines visual, textual, and spatial layout features
- Performs multimodal reasoning across tables and text
- Supports numerical reasoning over financial data
- Provides explainable answers with confidence scores

Training Strategy:

- Pretraining on general document datasets
- Fine-tuning on financial datasets for domain adaptation

Use Cases:

- Invoice understanding
- Financial report analysis
- Audit and compliance verification
- Intelligent document search
- Customer support QA systems

System Architecture

The proposed system follows a modular pipeline for financial document understanding and question answering.

Pipeline Stages:

- **Input Module:** Accepts financial documents (PDF/images)
- **Preprocessing:** Resizing, normalization, denoising, layout preservation
- **OCR + Layout Extraction:** Extract text, bounding boxes, table regions
- **Feature Encoder:** Combines visual, textual, and layout features
- **Vision-Language Transformer:** Multimodal reasoning across document
- **QA Module:** Interprets query and generates answers
- **Explainability Layer:** Highlights evidence + confidence score
- **Output Interface:** Displays answer with supporting evidence

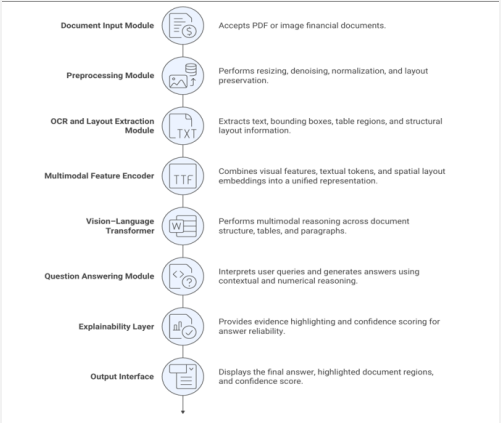

Figure 1.1: Financial Document VQA System Architecture

Layout-Aware Vision–Language DocVQA System Architecture

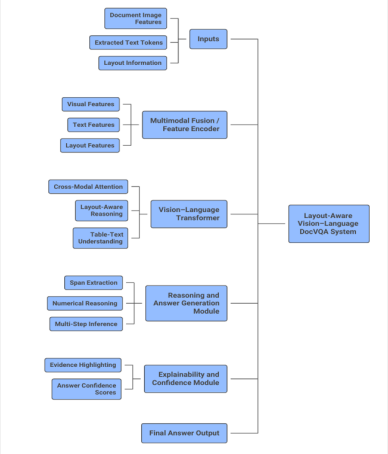

Figure 1.2: Layout-Aware Vision–Language DocVQA System Architecture

Dataset: TAT-DQA

TAT-DQA (Tabular And Textual Document Question Answering) is the primary financial DocVQA benchmark used in this project, consisting of 2,758 reports across 3,067 pages with 16,558 QA pairs.

Each sample includes tables (balance sheets, income statements) and narrative text, requiring:

- Revenue aggregation
- Growth calculation
- Region-based filtering
- Multi-step reasoning

Reasoning Distribution:

- ∼25% arithmetic (lookup + sum)
- ∼20% comparisons (max/min)
- ∼30% multi-step reasoning

Unlike FUNSD and DocVQA, TAT-DQA focuses on real financial reasoning tasks.

Pipeline: FUNSD → DocVQA → TAT-DQA

`github.com/NExTplusplus/TAT-DQA`

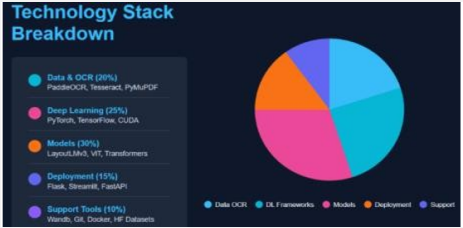

Figure 1.3: Sample Representation from TAT-DQA Dataset

Tools, Technologies & Protocols

Image Processing:

- PyMuPDF (fitz) – PDF → high-res images (300 DPI), preserves layout
- OpenCV – Resize, augment, denoise TAT-DQA pages
- Pillow (PIL) – Batch image loading/saving
- Albumentations – Rotation, blur for layout robustness

Layout & Table Parsing:

- Detectron2 – Table detection (Mask R-CNN trained on PubTabNet)
- Table Transformer – End-to-end table structure recognition
- Camelot/PyTabular – PDF table extraction (backup for non-image PDFs)

OCR & Text Extraction:

- PaddleOCR (PP-OCRv4) – Table detection + cell text + bbox (TAT-DQA best)
- Tesseract OCR – Fast word-level text + bounding boxes (FUNSD)
- EasyOCR – Simple setup, good accuracy baseline
- LayoutParser – Document segmentation (tables vs paragraphs)

Document Models:

- LayoutLMv3 – Text + bbox + image (microsoft/layoutlmv3-base)
- LayoutLMv2 – Text + bbox (FUNSD pretraining)
- Donut – OCR-free end-to-end (NAVER-Clova/donut-base)

Evaluation Metrics

Exact Match (EM): Measures whether the predicted answer exactly matches the ground truth.

$$EM = \begin{cases} 1, & \text{if Predicted} = \text{Ground Truth} \\ 0, & \text{otherwise} \end{cases}$$

$$EM\ Score = \frac{1}{N} \sum_{i=1}^{N} EM_i$$

F1-Score: Evaluates partial correctness using token overlap.

$$Precision = \frac{\text{Correct Tokens}}{\text{Predicted Tokens}}$$

$$Recall = \frac{\text{Correct Tokens}}{\text{Ground Truth Tokens}}$$

$$F1 = \frac{2 \times (Precision \times Recall)}{Precision + Recall}$$

OCR Accuracy: Measures quality of extracted text and its impact.

Processing Time: Measures efficiency for large-scale document processing.

$$\text{Avg Time} = \frac{\text{Total Processing Time}}{\text{Number of Documents}}$$

Answer Confidence Score: Indicates reliability of predictions.

$$\text{Confidence} = \max(P(a_1), P(a_2), \ldots, P(a_n))$$

Technical Validation

Validation: Ensure each model works independently and as a complete multi-model pipeline for financial document QA.

Component Validation:

- OCR $\rightarrow$ CER, WER
- Layout $\rightarrow$ IoU (bounding boxes)
- VQA $\rightarrow$ EM, F1-score

Integration Testing:

- Verify data flow between models
- Test on 100 documents

Model Comparison:

- PaddleOCR vs Tesseract
- LayoutLMv3 vs LayoutLMv2

Validation Activity

Validation Activity	Component	Metric
OCR Accuracy	PaddleOCR / Tesseract	CER, WER
Layout Extraction	Feature Extractor	IoU (bounding box)
VQA Answer Quality	LayoutLMv3 + ViT	EM, F1-Score
Confidence Calibration	Output Module	ECE
Inference Latency	Full Pipeline	Avg Time
Noise Robustness	Preprocessing + OCR	EM degradation

Table 1.6: Validation Activity Metrics

Dataset & Performance Validation

Dataset Validation:

- Data Split: 80% train / 10% val / 10% test
- Integrity Checks: Remove missing/invalid annotations
- OCR Verification: Manual check on 200 samples
- Generalisation: Tested on DocVQA and FinQA subsets

Performance Benchmarking:

- Baseline: LayoutLMv3 only
- SOTA Comparison: MHST, Doc2SoarGraph
- Ablation Study: Remove components and measure EM/F1 drop

Robustness & Reliability:

- Noise robustness (blur, rotation, compression)
- Layout variation testing
- Confidence calibration (ECE)
- Latency and throughput (GPU-based)

Expected Outcomes & Current Status

Expected Outcomes

- Accurate extraction of key financial information from document images and intelligent query answering.
- Reduced manual document review, leading to lower human errors and time savings.
- Improved handling of diverse document layouts compared to traditional OCR-based systems.
- Faster and more reliable processing of financial documents.
- Support for automation of financial workflows such as invoice processing, auditing, compliance, and information retrieval.

Current Status

- Problem statement and objectives of the DocVQA system have been finalized.
- Literature review on financial document–based QA systems completed.
- Relevant financial DocVQA dataset collected and confirmed.
- Dataset structure analyzed and overall system design planned.
- Project currently in the analysis and preparation stage implementation is yet to begin.

Project Timeline

Sem 6: Problem, survey, and baseline setup: Finalize problem statement, do detailed literature survey, select dataset (DOC-VQA), and build a simple baseline DocVQA pipeline.

Sem 7: Core model design and experimentation: Design and implement the integrated vision–language architecture (layout-aware transformer + OCR/layout stack), run main experiments on DOC-VQA, and evaluate using EM/F1 etc.

Sem 8: Optimization, deployment & thesis: Improve robustness, add explainability and confidence scores, prepare full documentation, deployment prototype, and final presentation.

Semester	Duration (approx.)	Main Objective	Key Outputs
6	Jan–May 2026	Define and ground the research problem in financial DocVQA	Problem statement, survey, dataset and tools finalized, baseline system
7	Jul–Nov 2026	Build and experiment with the proposed integrated vision–language model	Trained models, ablation results, quantitative evaluation, updated architecture
8	Jan–May 2027	Refine, optimize, and package the system for real-world use	Robust final model, deployment prototype, thesis, final viva/demo

Table 1.7: Project Timeline Overview

References

1. Kim et al., "Donut: OCR-Free Document Understanding Transformer," ECCV, 2022.
2. Xu et al., "LayoutLMv3: Unified Text and Image Masking," ACM MM, 2022.
3. Wang et al., "LaTr: Layout-Aware Transformer," CVPR, 2021.
4. Wu et al., "Survey on Visual Question Answering," IEEE TPAMI, 2023.
5. Li et al., "GenRecal," arXiv, 2025.
6. Reddy et al., "DocFinQA," arXiv, 2024.
7. Choi et al., "FinDER Dataset," arXiv, 2025.
8. Xue et al., "FAMMA Benchmark," arXiv, 2025.
9. Jin et al., "OCR + Vision-Language Models," arXiv, 2025.
10. Choi et al., "FinAgentBench," ACM AIFin, 2025.
11. Srivastava et al., "LLM Mathematical Reasoning in Financial DocVQA," arXiv, 2024.
12. Tao et al., "Multimodal Transformer QA," 2022.
13. Xue et al., "Retriever–Generator for Financial QA," 2021.
14. Mathew et al., "OCR-based Document QA," 2021.
15. Mathew et al., "Chart QA and Visual QA," 2022.
16. Zhu et al., "Financial Table QA," 2024.
17. Chen et al., "Multi-agent LLM Systems," 2025.
18. Xue et al., "Multimodal Financial QA," 2024.
19. Mathew et al., "Visual Document Understanding," 2021.
20. Liu et al., "Vision-Language Model Scaling," 2024.
21. Benchea et al., "DIVE-Doc," 2025.
22. Mohammadshirazi et al., "DLaVA," 2025.
23. Nguyen et al., "DocMIA," 2025.
24. Mohammadshirazi et al., "DocVAL," 2025.
25. Van Landeghem et al., "DUDE Benchmark," 2023.

Thank You

Questions?